

Theorem 5.11. *In the space $\mathbb{W}_p(\mathbb{R}^d)$, we have $W_p(\mu_n, \mu) \rightarrow 0$ if and only if $\mu_n \rightharpoonup \mu$ and $\int |x|^p d\mu_n \rightarrow \int |x|^p d\mu$.*

Proof. Consider first a sequence μ_n which is converging to μ in $\mathbb{W}_p(\mathbb{R}^d)$. It is still true in this case that

$$\sup \left\{ \int \phi d(\mu_n - \mu) : \phi \in \text{Lip}_1 \right\} \rightarrow 0,$$

which gives the convergence when testing against arbitrary Lipschitz functions. Note that Lipschitz functions are dense (for the uniform convergence) in the space $C_0(\Omega)$ (while it is not necessarily the case for $C_b(\Omega)$). This is enough to prove $\mu_n \rightarrow \mu$, thanks to Lemma 5.8.

To obtain the other condition, namely $\int |x|^p d\mu_n \rightarrow \int |x|^p d\mu$ (which is not a consequence of the weak convergence, since $|x|^p$ is not bounded), it is sufficient to note that

$$\int |x|^p d\mu_n = W_p^p(\mu_n, \delta_0) \rightarrow W_p^p(\mu, \delta_0) = \int |x|^p d\mu.$$

We need now to prove the opposite implication. Consider a sequence $\mu_n \rightharpoonup \mu$ satisfying also $\int |x|^p d\mu_n \rightarrow \int |x|^p d\mu$. Fix $R > 0$ and consider the function $\phi(x) := (|x| \wedge R)^p$, which is continuous and bounded. We have

$$\begin{aligned} \int (|x|^p - (|x| \wedge R)^p) d\mu_n &= \int |x|^p d\mu_n - \int \phi d\mu_n \\ &\rightarrow \int |x|^p d\mu - \int \phi d\mu = \int (|x|^p - (|x| \wedge R)^p) d\mu. \end{aligned}$$

Since $\int (|x|^p - (|x| \wedge R)^p) d\mu \leq \int_{B(0,R)^c} |x|^p d\mu$ and $\int_{\mathbb{R}^d} |x|^p d\mu < \infty$, it is possible to choose R so that

$$\int (|x|^p - (|x| \wedge R)^p) d\mu < \varepsilon/2$$

and hence one can also guarantee that $\int (|x|^p - (|x| \wedge R)^p) d\mu_n < \varepsilon$ for all n large enough.

We use now the inequality $(|x| - R)^p \leq |x|^p - R^p = |x|^p - (|x| \wedge R)^p$ which is valid for $|x| \geq R$ (see Lemma 5.12 below) to get

$$\int (|x| - R)^p d\mu_n < \varepsilon \text{ for } n \text{ large enough, and } \int (|x| - R)^p d\mu < \varepsilon.$$

Consider now $\pi_R : \mathbb{R}^d \rightarrow \overline{B(0,R)}$ defined as the projection over $\overline{B(0,R)}$. This map is well-defined and continuous and is the identity on $B(0,R)$. Moreover, for every $x \notin \overline{B(0,R)}$ we have $|x - \pi_R(x)| = |x| - R$. We can deduce

$$W_p(\mu, (\pi_R)_\# \mu) \leq \left(\int (|x| - R)^p d\mu \right)^{\frac{1}{p}} \leq \varepsilon^{\frac{1}{p}},$$

$$W_p(\mu_n, (\pi_R)_\# \mu_n) \leq \left(\int (|x| - R)^p d\mu_n \right)^{\frac{1}{p}} \leq \varepsilon^{\frac{1}{p}}.$$

Note also that, due to the usual composition of the weak convergence with continuous functions, from $\mu_n \rightharpoonup \mu$ we also infer $(\pi_R)_\# \mu_n \rightharpoonup (\pi_R)_\# \mu$. Yet, these measures are all concentrated on the compact set $\overline{B(0, R)}$ and here we can use the equivalence between weak convergence and W_p convergence. Hence, we get

$$\begin{aligned} \limsup_n W_p(\mu_n, \mu) &\leq \limsup_n (W_p(\mu_n, (\pi_R)_\# \mu_n) + W_p((\pi_R)_\# \mu_n, (\pi_R)_\# \mu) + W_p(\mu, (\pi_R)_\# \mu)) \\ &\leq 2\varepsilon^{\frac{1}{p}} + \lim_n W_p((\pi_R)_\# \mu_n, (\pi_R)_\# \mu) = 2\varepsilon^{\frac{1}{p}}. \end{aligned}$$

As the parameter $\varepsilon > 0$ is arbitrary, we get $\limsup_n W_p(\mu_n, \mu) = 0$ and the proof is concluded. $\square$

Lemma 5.12. *For $a, b \in \mathbb{R}_+$ and $p \geq 1$ we have $a^p + b^p \leq (a + b)^p$.*

Proof. Suppose without loss of generality that $a \geq b$. Then we can write $(a + b)^p = a^p + p\zeta^{p-1}b$, for a point $\zeta \in [a, a + b]$. Use now $p \geq 1$ and $\zeta \geq a \geq b$ to get $(a + b)^p \geq a^p + b^p$. $\square$

Remark 5.13. How to handle more general metric spaces X instead of subsets of $\mathbb{R}^d$? First, we notice that Lemma 5.8 does not hold in general (for instance, it is trivially false in infinite dimensional Banach spaces). Also, the problem of the density of Lipschitz functions in the space of continuous functions (even if X is compact) appears. Indeed, in $\mathbb{R}^d$ the fact that smooth functions are dense is easy to obtain by convolution, a tool which is not available in general spaces. To overcome this difficulty, one can prove the following statement, which is enough for our scopes: “if we have convergence testing against arbitrary Lipschitz functions, then we also have convergence testing against arbitrary functions in C_b ” (this is left as an exercise, see Ex(31)). Hence, in what concerns Theorems 5.9 and 5.10, everything can be translated into the framework of Polish spaces (with no need of local compactness). On the other hand the proof that we gave of Theorem 5.11 was somehow specific to $\mathbb{R}^d$: the key point was the existence of a continuous retraction π_R from the whole space to a compact subset (the ball $\overline{B(0, R)}$), with a bound on the distance between x and $\pi_R(x)$. The proof could be adapted to more general spaces, with ad-hoc assumptions, but we prefer to refer the reader to [17, 10] for the general case.

5.3 Lipschitz curves in $\mathbb{W}_p$ and the continuity equation

In this section we analyze some properties of Lipschitz and absolutely continuous curves in the space $\mathbb{W}_p$. In order to do that, we need some simple elements from analysis in metric spaces. The reader who wants to know more can have a look, for instance, at [13].

Box 5.1. – Good to know! – Curves and speed in metric spaces

Let us recall here some properties about Lipschitz curves in metric spaces.

A curve ω is a continuous function defined on a interval, say $[0, 1]$ and valued in a metric space (X, d) . As it is a map between metric spaces, it is meaningful to say whether it is Lipschitz or not, but its speed $\omega'(t)$ has no meaning, unless X is a vector space.

Surprisingly, it is possible to give a meaning to the modulus of the velocity, $|\omega'| (t)$.

Definition - If $\omega : [0, 1] \rightarrow X$ is a curve valued in the metric space (X, d) we define the metric derivative of ω at time t , denoted by $|\omega'| (t)$ through

$$|\omega'| (t) := \lim_{h \rightarrow 0} \frac{d(\omega(t+h), \omega(t))}{|h|},$$

provided this limit exists.

The following theorem, in the spirit of Rademacher Theorem (see Box 1.9 in Section 1.3) guarantees the existence of the metric derivative for Lipschitz curves.

Theorem - Suppose that $\omega : [0, 1] \rightarrow X$ is Lipschitz continuous. Then the metric derivative $|\omega'| (t)$ exists for a.e. $t \in [0, 1]$. Moreover we have, for $t < s$,

$$d(\omega(t), \omega(s)) \leq \int_t^s |\omega'| (\tau) d\tau.$$

The above theorem can be proved by using the fact that every compact metric space (and $\omega([0, 1])$ is the image of a compact set through a continuous map, hence it is compact) can be isometrically embedded in ℓ^∞ , where one can work componentwise. For all the notions and the proofs about metric derivatives, we refer for instance to [13], Chapter 4.

We also need to consider more general curves, not only Lipschitz continuous.

Definition - A curve $\omega : [0, 1] \rightarrow X$ is defined *absolutely continuous* whenever there exists $g \in L^1([0, 1])$ such that $d(\omega(t_0), \omega(t_1)) \leq \int_{t_0}^{t_1} g(s) ds$ for every $t_0 < t_1$. The set of absolutely continuous curves defined on $[0, 1]$ and valued in X is denoted by $AC(X)$.

It is well-known that every absolutely continuous curve can be reparametrized in time (through a monotone-increasing reparametrization) and become Lipschitz continuous. A possible way to achieve this goal is the following: let $G(t) := \int_0^t g(s) ds$, then set $S(t) = \varepsilon t + G(t)$ (for any $\varepsilon > 0$), which is continuous and strictly increasing, and valued in an interval $[0, L]$; for $t \in [0, L]$, set $\tilde{\omega}(t) = \omega(S^{-1}(t))$. It is easy to check that $\tilde{\omega} \in \text{Lip}_1$. Also, if we let $\varepsilon \rightarrow 0$ we obtain a unit-speed reparametrization of ω . If we want to have parametrizations defined on the same interval $[0, 1]$, we just need to rescale by a factor L .

In particular the above Rademacher Theorem is also true for $\omega \in AC(X)$ (since the reparametrization that we defined is differentiable a.e.).

The goal of this section is to identify the absolutely continuous curves in the space $\mathbb{W}_p(\Omega)$ with solutions of the continuity equation $\partial_t \mu_t + \nabla \cdot (\mathbf{v}_t \mu_t) = 0$ with L^p vector fields $\mathbf{v}_t$. Moreover, we want to connect the L^p norm of $\mathbf{v}_t$ with the metric derivative $|\mu'| (t)$.

We recall (see Section 4.1) that the continuity equation may be interpreted as the equation ruling the evolution of the density μ_t of a family of particles initially dis-

tributed according to μ_0 and each of which follows the flow

$$\begin{cases} y'_x(t) = \mathbf{v}_t(y_x(t)) \\ y_x(0) = x. \end{cases}$$

We state below the main theorem (originally proven in [17]) relating absolutely continuous curves in $\mathbb{W}_p$ with solutions of the continuity equation. For simplicity, we will only state it in the framework of a compact domain $\Omega \subset \mathbb{R}^d$. We will explain where we use the compactness assumption and how to get rid of it.

Theorem 5.14. *Let $(\mu_t)_{t \in [0,1]}$ be an absolutely continuous curve in $\mathbb{W}_p(\Omega)$ (for $p > 1$ and $\Omega \subset \mathbb{R}^d$ compact). Then for a.e. $t \in [0, 1]$ there exists a vector field $\mathbf{v}_t \in L^p(\mu_t; \mathbb{R}^d)$ such that*

- *the continuity equation $\partial_t \mu_t + \nabla \cdot (\mathbf{v}_t \mu_t) = 0$ is satisfied in the weak sense (see Section 4.1.2),*
- *for a.e. t we have $\|\mathbf{v}_t\|_{L^p(\mu_t)} \leq |\mu'|_t(t)$ (where $|\mu'|_t(t)$ denotes the metric derivative at time t of the curve $t \mapsto \mu_t$, w.r.t. the distance W_p);*

Conversely, if $(\mu_t)_{t \in [0,1]}$ is a family of measures in $\mathcal{P}_p(\Omega)$ and for each t we have a vector field $\mathbf{v}_t \in L^p(\mu_t; \mathbb{R}^d)$ with $\int_0^1 \|\mathbf{v}_t\|_{L^p(\mu_t)} dt < +\infty$ solving $\partial_t \mu_t + \nabla \cdot (\mathbf{v}_t \mu_t) = 0$, then $(\mu_t)_t$ is absolutely continuous in $\mathbb{W}_p(\Omega)$ and for a.e. t we have $|\mu'|_t(t) \leq \|\mathbf{v}_t\|_{L^p(\mu_t)}$.

Remark 5.15. Note that, as a consequence of the second part of the statement, the vector field $\mathbf{v}_t$ introduced in the first one must a posteriori satisfy $\|\mathbf{v}_t\|_{L^p(\mu_t)} = |\mu'|_t(t)$.

We will split the proof of this theorem into two parts, dealing with the direct and converse implications, respectively. We first introduce some tools that we need, and in particular the following one will be needed in both parts of the statement of Theorem 5.14.

Remark 5.16. It is important to observe that, by reparametrizing in time, it is always possible to assume that the absolutely continuous curve μ_t of the first part of the statement is actually Lipschitz (and we obtain in this case a uniform bound on $\|\mathbf{v}_t\|_{L^p(\mu_t)}$), and that the norm $\|\mathbf{v}_t\|_{L^p(\mu_t)}$ in the second part is bounded (and we obtain that μ_t is Lipschitz). Hence, all the proofs will be done in this Lipschitz framework.

5.3.1 The Benamou-Brenier functional $\mathcal{B}_p$

We need to start from an easy computation, which will be useful in the sequel. We consider a pair of exponents p and q such that $\frac{1}{p} + \frac{1}{q} = 1$.

Lemma 5.17. *Set $K_q := \{(a, b) \in \mathbb{R} \times \mathbb{R}^d : a + \frac{1}{q}|b|^q \leq 0\}$. Then, for $(t, x) \in \mathbb{R} \times \mathbb{R}^d$ we have*

$$\sup_{(a,b) \in K_q} (at + b \cdot x) = f_p(t, x) := \begin{cases} \frac{1}{p} \frac{|x|^p}{t^{p-1}} & \text{if } t > 0, \\ 0 & \text{if } t = 0, x = 0 \\ +\infty & \text{if } t = 0, x \neq 0, \text{ or } t < 0. \end{cases} \quad (5.2)$$

In particular f_p is convex and l.s.c..

Proof. First of all, we establish (5.2). Suppose $t > 0$: then it is clear that one should take the maximal possible value of a in the sup, and hence $a = -\frac{1}{q}|b|^q$. This gives

$$\sup_b \left(-\frac{1}{q}|b|^q + b \cdot x \right) = t \left(\sup_b -\frac{1}{q}|b|^q + b \cdot \left(\frac{x}{t}\right) \right).$$

If we recall that the Legendre transform of $p \mapsto \frac{1}{q}|p|^q$ is $x \mapsto \frac{1}{p}|x|^p$. This gives

$$\sup_b -\frac{1}{q}|b|^q + b \cdot y = \frac{1}{p}|y|^p \quad \text{for all } y, \quad (5.3)$$

and hence

$$\sup \left\{ at + b \cdot x : a \in \mathbb{R}, b \in \mathbb{R}^d, a + \frac{1}{q}|b|^q \leq 0 \right\} = t \frac{1}{p} \left| \frac{x}{t} \right|^p = \frac{1}{p} \frac{|x|^p}{t^{p-1}}.$$

The case $(t, x) = (0, 0)$ is straightforward. If $t = 0$ and $x \neq 0$ then it is clear that any vector b may be compensated by a sufficiently negative value of a , which gives

$$\sup \left\{ at + b \cdot x : a \in \mathbb{R}, b \in \mathbb{R}^d, a + \frac{1}{q}|b|^q \leq 0 \right\} = \sup_b b \cdot x = +\infty.$$

Finally, in the case $t < 0$ one can take a arbitrarily negative and $b = 0$ so that

$$\sup \left\{ at + b \cdot x : a \in \mathbb{R}, b \in \mathbb{R}^d, a + \frac{1}{q}|b|^q \leq 0 \right\} \geq \sup_{a < 0} at = +\infty.$$

The fact that f_p is convex and l.s.c. is a consequence of the fact that it is expressed as a supremum of linear functions. $\square$ $\square$

We can note in the above result that we computed the Legendre transform of an indicator function, and that we got, as expected, a 1-homogeneous convex function. We want now to use this function f_p to define a functional over measures. For $\rho \in \mathcal{M}(X)$ and $E \in \mathcal{M}^d(X)$ we set

$$\mathcal{B}_p(\rho, E) := \sup \left\{ \int_X a(x) d\rho + \int_X b(x) \cdot dE : (a, b) \in C_b(X; K_q) \right\},$$

where the letter $\mathcal{B}$ is chosen because of the role that this functional has in the so-called Benamou-Brenier formula (see Section 6.1).

We prove the following

Proposition 5.18. *The functional $\mathcal{B}_p$ is convex and lower semi-continuous on the space $\mathcal{M}(X) \times \mathcal{M}^d(X)$ for the weak convergence. Moreover the following properties hold*

1. $\mathcal{B}_p(\rho, E) \geq 0$,

2. $\mathcal{B}_p(\rho, E) := \sup\{\int a(x)d\rho + \int b(x) \cdot dE : (a, b) \in L^\infty(X; K_q)\},$
3. if both ρ and E are absolutely continuous w.r.t. a same positive measure λ on X , we can write $\mathcal{B}_p(\rho, E) = \int_X f_p(\rho(x), E(x)) d\lambda(x)$, where we identify ρ and E with their densities w.r.t. λ ,
4. $\mathcal{B}_p(\rho, E) < +\infty$ only if $\rho \geq 0$ and $E \ll \rho$,
5. for $\rho \geq 0$ and $E \ll \rho$, we have $E = \mathbf{v} \cdot \rho$ and $\mathcal{B}_p(\rho, E) = \int \frac{1}{p} |\mathbf{v}|^p d\rho$,
6. If $X = \mathbb{R}^d$, $\rho^\varepsilon := \rho * \eta_\varepsilon$ and $E^\varepsilon := E * \eta_\varepsilon$ (for standard even mollifying kernel η_ε), then we have $\mathcal{B}_p(\rho^\varepsilon, E^\varepsilon) \leq \mathcal{B}_p(\rho, E)$.

Proof. The convexity and lower semi-continuity of $\mathcal{B}_p$ are a direct consequence of its definition as a supremum of linear functionals. The first property (positivity) is straightforward, as one can take $a = 0$ and $b = 0$ in the supremum.

In order to prove that the supremum can be taken over bounded functions instead of continuous ones we just need to approximate every bounded function with continuous ones (beware that, by abuse of notation, the space $L^\infty(X; K_q)$ denotes here the space of bounded measurable functions valued in K_q , independently of the reference measure). This can be done via Lusin's Theorem (see the corresponding Box 1.6 in Section 1.1), and every bounded pair (a, b) can be approximated by continuous pairs $(\tilde{a}, \tilde{b})$, such that $(\rho + |E|)(\{(a, b) \neq (\tilde{a}, \tilde{b})\}) < \varepsilon$ and $\sup |\tilde{a}| \leq \sup |a|$ and $\sup |\tilde{b}| \leq \sup |b|$. Then, we can replace $\tilde{a}$ with $\min\{\tilde{a}, -\frac{1}{q}|\tilde{b}|^q\}$ so as to guarantee to have functions valued in K_q . The difference $\left| \int a(x)d\rho + \int b(x) \cdot dE - \int \tilde{a}(x)d\rho + \int \tilde{b}(x) \cdot dE \right|$ is bounded by $C\varepsilon$ where C only depends on $\sup |a|$ and $\sup |b|$.

Once we know that we can use measurable functions a, b with no need of continuity, then Property 3. is a consequence of Lemma 5.2. As a consequence of Lemma 5.17 we can write

$$\begin{aligned} \mathcal{B}_p(\rho, E) &= \sup \left\{ \int [a(x)\rho(x) + b(x) \cdot E(x)] d\lambda(x) : (a, b) \in L^\infty(X; K_q) \right\} \\ &= \int f_p(\rho(x), E(x)) d\lambda(x). \end{aligned}$$

The fact that $\mathcal{B}_p$ is only finite if $\rho \geq 0$ is easy: assume it to be false for contradiction, and take a set A such that $\rho(A) < 0$, $a = -n\mathbb{1}_A$ and $b = 0$. We have $\mathcal{B}_p(\rho, E) \geq n\rho(A)$ and, since n is arbitrary, $\mathcal{B}_p(\rho, E) = +\infty$. To prove the fact that $\mathcal{B}_p$ is only finite if $E \ll \rho$, assume that there is a set A with $\rho(A) = 0$ and $E(A) \neq 0$. Take any unit vector e and use $a = -\frac{n^q}{q}\mathbb{1}_A$ and $b = ne\mathbb{1}_A$. We have $\mathcal{B}_p(\rho, E) \geq ne \cdot E(A)$. Since both e and n are arbitrary, we get $\mathcal{B}_p(\rho, E) = +\infty$.

When we restrict to the case $\rho \geq 0$ and $E = \mathbf{v} \cdot \rho$, property 5. is a consequence of property 3. Indeed, just use $\lambda = \rho$, thus getting

$$\mathcal{B}_p(\rho, \mathbf{v} \cdot \rho) = \int f_p(1, \mathbf{v}(x)) d\rho(x) = \int \frac{1}{p} |\mathbf{v}|^p d\rho.$$

Let us now prove property 6. Take arbitrary bounded functions $a : \mathbb{R}^d \rightarrow \mathbb{R}$ and $b : \mathbb{R}^d \rightarrow \mathbb{R}^d$ satisfying $a(x) + \frac{1}{q}|b(x)|^q \leq 0$ for all x . From standard properties of convolutions we have

$$\int a \, d\rho^\varepsilon + \int b \cdot dE^\varepsilon = \int a^\varepsilon \, d\rho + \int b^\varepsilon \cdot dE,$$

where $a^\varepsilon := a * \eta_\varepsilon$ and $b^\varepsilon := b * \eta_\varepsilon$. Note that, by Jensen inequality

$$|b^\varepsilon(y)|^q = \left| \int b(x) \eta_\varepsilon(x-y) \, dx \right|^q \leq \int |b(x)|^q \eta_\varepsilon(x-y) \, dx$$

and hence

$$a^\varepsilon(y) + \frac{1}{q}|b^\varepsilon(y)|^q \leq \int a(x) + \left(\frac{1}{q}|b(x)|^q \right) \eta_\varepsilon(x-y) \, dx \leq 0.$$

This proves $(a^\varepsilon, b^\varepsilon) \in C(\mathbb{R}^d; K_q)$ and hence we have

$$\int a \, d\rho^\varepsilon + \int b \cdot dE^\varepsilon = \int a^\varepsilon \, d\mu + \int b^\varepsilon \cdot dE \leq \mathcal{B}_p(\rho, E).$$

Passing to the sup in a and b , one gets $\mathcal{B}_p(\rho^\varepsilon, E^\varepsilon) \leq \mathcal{B}_p(\rho, E)$. $\square$ $\square$

Remark 5.19. As a convex and l.s.c. functional, $\mathcal{B}_p$ has admits a subdifferential, which one could usefully compute for optimization purposes: this computation is developed in [235] but we will not need it in this framework.

We are now ready to prove the first part of Theorem 5.14.

5.3.2 Proof of Theorem 5.14 - Part 1: μ is AC $\Rightarrow$ there exists v

Proof. First of all we note that it is not restrictive to assume that μ is a Lipschitz curve (if not, we just reparametrize in time).

We will do our proof by approximation, i.e. we build a sequence μ_t^k of curves which admit the existence of a suitable vector field v_t^k , in such a way that $\mu_t^k \rightarrow \mu_t$ and that we can find a limit for the vector fields also.

To define μ_t^k we proceed in the following way. For $t = i/k, i = 0, 1, \dots, k$, we define $\mu_t^k = \mu_t * \eta_k$, where η_k is a smooth approximation of the identity, i.e. a smooth even positive function supported on $B(0, 1/k)$ and with integral equal to 1. The only goal of this smoothing is to guarantee that the measures are absolutely continuous. The support of the measures is now contained in a slightly bigger domain $\Omega' \supset \Omega$, that we can choose convex. Then, for each $i = 0, 1, \dots, k-1$, we define $T^{i,k} : \Omega' \rightarrow \Omega'$ to be the optimal transport (for the cost $c(x, y) = |x - y|^p$) from $\mu_{i/k}^k$ to $\mu_{(i+1)/k}^k$. These optimal transport maps exist since every $\mu_{i/k}^k$ is absolutely continuous. Then, for $t \in]\frac{i}{k}, \frac{i+1}{k}[$, we define

$$\mu_t^k := ((i+1-kt)\text{id} + (kt-i)T^{i,k})_{\#} \mu_{i/k}^k,$$

which means that the particle located at x at time i/k goes to the position $T^{i,k}(x)$ at time $(i+1)/k$, moving with constant speed on the segment connecting these two positions

in the interval $]\frac{i}{k}, \frac{i+1}{k}[$. The velocity of this movement is given by $\mathbf{v}^{i,k} := k(\mathbf{T}^{i,k}(x) - x)$. Set $\mathbf{T}_t^{i,k} := (i+1-kt)\text{id} + (kt-i)\mathbf{T}^{i,k}$; this map is injective (see Lemma 4.23). Thus, we can set $\mathbf{v}_t^k := \mathbf{v}^{i,k} \circ (\mathbf{T}_t^{i,k})^{-1}$. This last vector field exactly represents the velocity that each particle has during its movement at time t : first we detect which is the original position of such a particle (at time i/k), then we associate to this particle the velocity associated to this original position.

As a consequence of this construction and of the considerations in Section 4.1, the pair $(\mu_t^k, \mathbf{v}_t^k)$ solves the continuity equation $\partial_t \mu_t^k + \nabla \cdot (\mathbf{v}_t^k \mu_t^k) = 0$.

The important point is the computation

$$\begin{aligned} \|\mathbf{v}_t^k\|_{L^p(\mu_t^k)}^p &= \int |\mathbf{v}_t^k|^p d\mu_t^k = \int |\mathbf{v}^{i,k}|^p \circ (\mathbf{T}_t^{i,k})^{-1} d(\mathbf{T}_t^{i,k})_{\#} \mu_{i/k}^k = \int |\mathbf{v}^{i,k}|^p d\mu_{i/k}^k \\ &= k^p W_p^p(\mu_{i/k}^k, \mu_{(i+1)/k}^k) \leq k^p W_p^p(\mu_{i/k}, \mu_{(i+1)/k}) \\ &\leq k^p \left(\int_{i/k}^{(i+1)/k} |\mu'|^p(s) ds \right)^p \leq k \int_{i/k}^{(i+1)/k} |\mu'|^p(s) ds. \end{aligned}$$

Here we used the fact that $W_p(\mu_{i/k} * \eta_\varepsilon, \mu_{(i+1)/k} * \eta_\varepsilon) \leq W_p(\mu_{i/k}, \mu_{(i+1)/k})$ (see Lemma 5.2) and, at the end, Jensen's inequality. In particular, we also get a uniform bound on $\|\mathbf{v}_t^k\|_{L^p(\mu_t^k)}$ since $k \int_{i/k}^{(i+1)/k} |\mu'|^p(s) ds \leq \text{Lip}(\mu)$.

This implies that, for arbitrary time $0 \leq a < b \leq 1$ (with $i_a/k \leq a \leq (i_a+1)/k$ and $i_b/k \leq b \leq (i_b+1)/k$), we have

$$\begin{aligned} \int_a^b \|\mathbf{v}_t^k\|_{L^p(\mu_t^k)}^p dt &\leq \sum_{i=i_a}^{i_b} \int_{i/k}^{(i+1)/k} |\mu'|^p(s) ds \\ &= \int_{i_a/k}^{(i_b+1)/k} |\mu'|^p(s) ds \leq \int_a^b |\mu'|^p(s) ds + \frac{2\text{Lip}(\mu)}{k}. \end{aligned} \quad (5.4)$$

Now, define a vector measure E^k on $\Omega' \times [0, 1]$ via

$$\int \phi(t, x) \cdot dE^k := \int_0^1 \left(\int \phi(t, x) \cdot \mathbf{v}_t^k(x) d\mu_t^k(x) \right) dt.$$

We can easily check that

$$\|E^k\| = \int_0^1 \|\mathbf{v}_t^k\|_{L^1(\mu_t^k)} dt \leq \int_0^1 \|\mathbf{v}_t^k\|_{L^p(\mu_t^k)} dt \leq \left(\int_0^1 \|\mathbf{v}_t^k\|_{L^p(\mu_t^k)}^p dt \right)^{\frac{1}{p}} \leq C.$$

Hence, up to a subsequence extraction, the sequence of vector measures E^k admits a weak limit E . On the other hand, it is easy to check that μ^k converges in $\mathbb{W}_p$, uniformly in time, to μ . Indeed

$$W_p(\mu_t^k, \mu_t) \leq W_p(\mu_t^k, \mu_{i/k}^k) + W_p(\mu_{i/k}^k, \mu_{i/k}) + W_p(\mu_{i/k}, \mu_t).$$

The first term may be estimated using

$$W_p(\mu_t^k, \mu_{i/k}^k) \leq |t - i/k| \|\mathbf{v}^{i,k}\|_{L^p(\mu_{i/k}^k)} \leq \frac{C}{k};$$

the second may be estimated by the size of the support of the kernel η_ε (which allows to obtain $\mu_{i/k}^k$ from $\mu_{i/k}$), which is $\varepsilon = 1/k$; the third is estimated by the Lipschitz constant of μ times, again, $|t - \frac{i}{k}| \leq \frac{1}{k}$. Overall, we get

$$W_p(\mu_t^k, \mu_t) \leq \frac{C}{k}.$$

From $\partial_t \mu^k + \nabla \cdot E^k = 0$, we can easily pass to the limit and get $\partial_t \mu + \nabla \cdot E = 0$. We want now to prove that $E_t = \mathbf{v}_t \mu_t$, with $\|\mathbf{v}_t\|_{L^p(\mu_t)} \leq |\mu'|^p(t)$ a.e.

This is a consequence of Proposition 5.18 on the semi-continuity of $\mathcal{B}_p$. Indeed, using (5.4) with $a = 0, b = 1$, we have $\mathcal{B}_p(\mu^k, E^k) \leq C$ (where the functional $\mathcal{B}_p$ is considered on the whole space-time domain $[0, 1] \times \Omega$). This is enough to obtain $\mathcal{B}_p(\mu, E) < +\infty$ and $E \ll \mu$, i.e. $E_t = \mathbf{v}_t \mu_t$. Then, if we apply the same lemma to the measures μ^k and E^k restricted to a closed interval $[a, b]$ we obtain

$$\int_a^b \|\mathbf{v}_t\|_{L^p(\mu_t)}^p dt \leq \liminf_k \int_a^b \|\mathbf{v}_t^k\|_{L^p(\mu_t^k)}^p dt \leq \int_a^b |\mu'|^p(t) dt.$$

Since this is true for arbitrary values of a and b , on every Lebesgue point t_0 of $t \mapsto \|\mathbf{v}_t\|_{L^p(\mu_t)}^p$ and $t \mapsto |\mu'|^p(t)$ we get the desired inequality. $\square$ $\square$

We want now to prove the converse implication of Theorem 5.14. Again, to be able to perform our computations rigorously, we need to regularize by convolution. The construction that we do here will be useful many times, and it deserves to be detailed and analyzed.

5.3.3 Regularization of solutions of the continuity equation.

Suppose that (μ, E) solves the continuity equation $\partial_t \mu_t + \nabla \cdot E_t = 0$, that the supports of all measures μ_t and E_t are contained in a same compact set $\Omega \subset \mathbb{R}^d$, and that E_t is bounded in the space of measure, i.e. $\|E_t\| := |E_t|(\mathbb{R}^d) \leq C$. Take a C^∞ strictly positive kernel η_ε (as a function of space only) and define $\mu_t^\varepsilon := \eta_\varepsilon * \mu_t$ and $E_t^\varepsilon := \eta_\varepsilon * E_t$, and $\mathbf{v}_t^\varepsilon := E_t^\varepsilon / \mu_t^\varepsilon$. The choice is made so that $\partial_t \mu_t^\varepsilon + \nabla \cdot (\mu_t^\varepsilon \mathbf{v}_t^\varepsilon) = 0$.

Since we want to apply the characterization result of solutions of the continuity equation (Theorem 4.4), we want to check uniform global bounds on $\mathbf{v}^\varepsilon$. To do this, we need to perform a particular choice of η_ε , and in particular we take of the form $\eta^\varepsilon(z) = \varepsilon^{-d} \eta(|z|/\varepsilon)$, where $\eta : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ is a smooth function such that $\eta(t) = Ce^{-t}$ for $t \geq 1$. In Lemma 5.20 below, we check that this guarantees that $\mathbf{v}^\varepsilon$ satisfies the required global bounds, and we also state regularity for μ_ε (that we will need in Section 5.3.5).

Lemma 5.20. *With the above choice of the convolution kernel η^ε , the function μ^ε is Lipschitz in (t, x) and $\mathbf{v}^\varepsilon$ is Lipschitz in x and bounded, uniformly in t (for fixed $\varepsilon > 0$), provided Ω is bounded.*

Proof. We can fix $\varepsilon = 1$ as the computations for other values of ε are similar. We set $\eta = \eta^\varepsilon$. From $\mu^\varepsilon(t, x) = \int \eta(|y - x|) \mu_t(dy)$ we have a Lipschitz bound in x from

standard convolution properties, but also

$$\partial_t \mu^\varepsilon(t, x) = \frac{d}{dt} \int \eta(y - x) d\mu_t(x) = \int \nabla \eta(y - x) \cdot dE_t(x),$$

which is bounded by $\text{Lip}(\eta)|E_t|(\mathbb{R}^d)$.

As far as $\mathbf{v}^\varepsilon$ is concerned, computations are trickier since it is defined as $E^\varepsilon/\mu^\varepsilon$. From $\eta > 0$ we get $\mu^\varepsilon > 0$ which guarantees that $\mathbf{v}^\varepsilon$ is smooth as the ratio between two smooth functions, with non-vanishing denominator. Yet, we want to compute explicit and global bounds for large $|x|$. If $R_0 = \text{diam}(\Omega)$ and $0 \in \Omega$ we have for sure, for $|x| = R \geq 1$, $\mu^\varepsilon(x) \geq \eta(R + R_0)$. On the other hand we have, for $R \geq 2R_0 \vee 1$, $|E^\varepsilon| \leq C\eta(R - R_0)$, $|\nabla E^\varepsilon| \leq C|\eta'(R - R_0)|$, and, similarly, $|\nabla \mu^\varepsilon| \leq C|\eta'(R - R_0)|$. From

$$|\mathbf{v}^\varepsilon| = \frac{|E^\varepsilon|}{\mu^\varepsilon}, \quad |\nabla \mathbf{v}^\varepsilon| \leq \frac{|\nabla E^\varepsilon|}{\mu^\varepsilon} + \frac{|E^\varepsilon| |\nabla \mu^\varepsilon|}{(\mu^\varepsilon)^2},$$

using $\eta = \eta' = \eta''$ on $[1, +\infty[$ and $\eta(R + R_0) = e^{-2R_0} \eta(R - R_0)$, we get uniform bounds. $\square$

We will see that the application of the above Lemma 5.20 is the only point where we needed Ω to be bounded. Without this assumption, we cannot apply the characterization result for the continuity equation that we presented in Theorem 4.4. On the other hand, we already mentioned that more general uniqueness results exist (for instance in [17]), and we simply chose not to present them for the sake of simplicity.

Thanks to our definitions, we have

$$\partial_t \mu_t^\varepsilon + \nabla \cdot (\mathbf{v}_t^\varepsilon \mu_t^\varepsilon) = 0.$$

Moreover, applying Theorem 4.4, we know that the only solution of $\partial_t \rho_t + \nabla \cdot (\mathbf{v}_t^\varepsilon \rho_t) = 0$ is given by the flow of $\mathbf{v}^\varepsilon$. Hence, we get

$$\mu_t^\varepsilon = (Y_t)_\# \mu_0^\varepsilon, \quad \text{where } Y_t(x) = y_x(t), \quad \text{and } \begin{cases} y'_x(t) = \mathbf{v}_t^\varepsilon(y_x(t)), \\ y_x(0) = x. \end{cases} \quad (5.5)$$

We can summarize our construction in the following proposition.

Proposition 5.21. *Consider a pair (μ, E) solving $\partial_t \mu_t + \nabla \cdot E_t = 0$, and suppose that the supports of all measures μ_t and E_t are contained in a same compact set $\Omega \subset \mathbb{R}^d$, and that E_t is bounded in the space of measure, i.e. $|E_t|(\mathbb{R}^d) \leq C$. Then, there exists a family of approximating curves $(\mu_t^\varepsilon)_t$ with a velocity vector field $\mathbf{v}_t^\varepsilon$ such that*

- μ^ε is Lipschitz in (t, x) and $\mathbf{v}^\varepsilon$ is Lipschitz in x and bounded, uniformly in t ; moreover, $\mu^\varepsilon > 0$ everywhere;
- μ_t^ε is obtained from μ_0^ε by following the flow of the vector field $\mathbf{v}^\varepsilon$, in the sense of (5.5);
- For every t we have $\mu_t^\varepsilon \rightharpoonup \mu_t$ and for a.e. t we have $E_t^\varepsilon \rightharpoonup E_t$.
- $\|\mathbf{v}_t^\varepsilon\|_{L^p(\mu_t^\varepsilon)} \leq \mathcal{B}_p(\mu_t, E_t)$.

Proof. We use the construction we described above. The first property is proven in Lemma 5.20, the second has been discussed above, and the third is evident by construction. The fourth is a consequence of Property 5. of Lemma 5.18. Note that this means $\|\mathbf{v}_t^\varepsilon\|_{L^p(\mu_t^\varepsilon)} \leq \|\mathbf{v}_t\|_{L^p(\mu_t)}$ whenever $E_t = \mathbf{v}_t \cdot \mu_t$. $\square$ $\square$

We finish this paragraph by noting that the same construction could have been performed on the torus.

Remark 5.22. If $\mu_t \in \mathcal{P}(\mathbb{T}^d)$, one can consider an arbitrary approximation kernel $\eta^\varepsilon \in C^\infty(\mathbb{R}^d)$ of the form $\eta^\varepsilon(z) = \varepsilon^{-d} \eta(z/\varepsilon)$ (strictly positive, but with no need of imposing exponential behavior) and define the convolution on the torus. To be precise, let us stress that the periodic convolution $\eta * \mu$ between a measure μ on $\mathbb{T}^d$ and a function η on $\mathbb{R}^d$ is defined as follows. Consider at first μ as a measure on $Q = [-\frac{1}{2}, \frac{1}{2}] \subset \mathbb{R}^d$, perform standard convolution on $\mathbb{R}^d$, and then define

$$(\eta * \mu)(x) := \sum_{k \in \mathbb{Z}^d} (\eta * \mu)(x + k).$$

Equivalently, this corresponds to a convolution on the torus with the convolution kernel $\sum_{k \in \mathbb{Z}^d} \eta(x + k)$.

With this construction, the same conclusions as in Proposition 5.21 hold. Moreover, every compact domain Ω can be seen as a subset of a cube Q identified with the torus, and it can even fit an arbitrary small portion of Q . In this case, every optimal transport problem for the costs $|x - y|^p$ between measures on Ω “does not see” the fact that Q is identified with a torus (it is enough to have $\Omega \subset \frac{1}{2}Q$).

We stress that the uniform bounds in the case of the torus are easier to obtain, but we found conceptually easier to deal with measures on $\mathbb{R}^d$ (and in particular with optimal transport problems in $\mathbb{R}^d$). This is why we prefer the Euclidean construction to this one. Yet, we present both of them for the use that we will do in Section 5.3.5, where compactness is crucial for our estimates.

5.3.4 Proof of Theorem 5.14 - Part 2: there exists $\mathbf{v} \Rightarrow \mu$ is AC

Proof. The starting point is a pair (μ, E) which solves the continuity equation, where $E_t = \mathbf{v}_t \cdot \mu_t$ and $\mathbf{v}_t \in L^p(\mu - t)$ for a.e. t . Again, we apply a reparametrization: with a change of variable in time we can replace (μ_t, E_t) with a new pair where E_t is just multiplied by a scalar constant depending on t . By choosing our change of variable, we can impose a uniform bound on $\|\mathbf{v}_t\|_{L^p(\mu_t)}$. This also implies a bound on $\|E_t\| = |E_t|(\mathbb{R}^d) = \|\mathbf{v}_t\|_{L^1(\mu_t)}$.

Under this extra assumption (the uniform bound on $\|E_t\|$), we can use the construction provided in the previous paragraph (using the construction on $\mathbb{R}^d$ for simplicity of exposition). Thus, we can build an approximated curve μ_t^ε which is driven by the flow of the smooth (in space) vector field $\mathbf{v}^\varepsilon$. This provides a useful transport plan between

μ_t^ε and μ_{t+h}^ε , taking $\gamma = (T_t, T_{t+h})_\# \mu_0^\varepsilon \in \Pi(\mu_t^\varepsilon, \mu_{t+h}^\varepsilon)$. We obtain

$$\begin{aligned}
 W_p(\mu_t^\varepsilon, \mu_{t+h}^\varepsilon) &\leq \left(\int_{\Omega \times \Omega} |x - y|^p d\gamma \right)^{\frac{1}{p}} = \left(\int_{\Omega} |T_t(x) - T_{t+h}(x)|^p d\mu_0^\varepsilon \right)^{\frac{1}{p}} \\
 &\leq |h|^{1/q} \left(\int_{\Omega} \int_t^{t+h} \left| \frac{d}{ds} T_s(x) \right|^p ds d\mu_0^\varepsilon \right)^{\frac{1}{p}} \\
 &= |h|^{1/q} \left(\int_t^{t+h} ds \int_{\Omega} |\mathbf{v}_s^\varepsilon(y_s(s))|^p d\mu_0^\varepsilon \right)^{\frac{1}{p}} \\
 &= |h|^{1/q} \left(\int_t^{t+h} ds \int_{\Omega} |\mathbf{v}_s^\varepsilon(y)|^p d\mu_s^\varepsilon(y) \right)^{\frac{1}{p}}.
 \end{aligned}$$

We proved in Lemma 5.17 that $\int |\mathbf{v}_s^\varepsilon(y)|^p d\mu_s^\varepsilon(y) \leq \|\mathbf{v}_s\|_{L^p(\mu_s)}^p$. From this, we get

$$W_p(\mu_t^\varepsilon, \mu_{t+h}^\varepsilon) \leq |h|^{1/q} \left(\int_t^{t+h} \|\mathbf{v}_s\|_{L^p(\mu_s)}^p ds \right)^{\frac{1}{p}} = |h| \left(\frac{1}{|h|} \int_t^{t+h} \|\mathbf{v}_s\|_{L^p(\mu_s)}^p ds \right)^{\frac{1}{p}}.$$

First we can pass to the limit $\varepsilon \rightarrow 0$, thus obtaining

$$\frac{W_p(\mu_t, \mu_{t+h})}{|h|} \leq \left(\frac{1}{|h|} \int_t^{t+h} \|\mathbf{v}_s\|_{L^p(\mu_s)}^p ds \right)^{\frac{1}{p}}$$

and then as $h \rightarrow 0$ at points t which are Lebesgue points of $t \mapsto \|\mathbf{v}_t\|_{L^p(\mu_t)}^p$, we obtain

$$|\mu'|_t(t) \leq \|\mathbf{v}_t\|_{L^p(\mu_t)},$$

which provides the desired estimate. $\square$ $\square$

We remark that Theorem 5.14 has been proven thanks to an extensive use of the functional $\mathcal{B}_p$: we needed its convexity in the second part and its semi-continuity in the first part⁵.

Remark 5.23. We also note that we only defined the functional $\mathcal{B}_p$ for $p > 1$, and that the only natural extension for $p = 1$ would be given by $\mathcal{B}_1(\rho, E) = \|E\|$. It is easy to understand that the first part of Theorem 5.14 cannot stay true in the case $p = 1$. Indeed, the part of the statement which fails is the existence of a vector field $\mathbf{v}_t$, since it is no more possible to deduce $E \ll \mu$: it is not difficult to adapt the proof so as to get the existence of a family E_t of vector measures solving $\partial_t \mu_t + \nabla \cdot E_t = 0$, but E_t

⁵The proof that we gave of the second part is standard, and based on the interpretation of the continuity equation that we gave in Chapter 4. The approximations that we performed are the same as in [17], but we tried to simplify them to make this exposition self-contained: in order to do that, we chose a very precise convolution kernel. Concerning the first part, we stress that a different, and much more elegant, approach can be found in [17] and does not use approximation. The approach that we presented here is more or less inspired by a paper by Lisini, [213].

will be no more of the form $\mathbf{v}_t \cdot \mu_t$. An example where absolute continuity fails is the following: take $\mu_t := (1-t)\delta_{x_0} + t\delta_{x_1}$, which is Lipschitz in $\mathbb{W}_1$ (but not in $\mathbb{W}_p$, $p > 1$). In this case one could take $E_t = \frac{x_1 - x_0}{|x_1 - x_0|} \mathcal{H}^1 \llcorner [x_0, x_1]$ (a uniform measure on the segment joining the two points, oriented as the same segment), but no measure concentrated on $\{x_0, x_1\}$ can be such that its divergence is $\delta_{x_1} - \delta_{x_0}$.

5.3.5 Derivative of W_p^P along curves of measures

We conclude this section with a useful computation that can be applied in many cases in the study of evolution PDEs. The same result is efficiently proven in [17] but we provide a different proof.

Theorem 5.24. *Let $(\rho_t^{(i)}, \mathbf{v}_t^{(i)})$ for $i = 1, 2$ be two solutions of the continuity equation $\partial_t \rho_t^{(i)} + \nabla \cdot (\mathbf{v}_t^{(i)} \rho_t^{(i)}) = 0$ on a compact domain Ω (with homogeneous Neumann boundary conditions) and suppose that $\rho_t^{(i)} \ll \mathcal{L}^d$ for every t and that $\rho^{(i)}$ are absolutely continuous curves in $\mathbb{W}_p(\Omega)$. Then we have,*

$$\frac{d}{dt} \left(\frac{1}{p} W_p^p(\rho_t^{(1)}, \rho_t^{(2)}) \right) = \int \nabla \varphi_t \cdot \mathbf{v}_t^{(1)} \rho_t^{(1)} dx + \int \nabla \psi_t \cdot \mathbf{v}_t^{(2)} \rho_t^{(2)} dx$$

for a.e. t , where (φ_t, ψ_t) is any pair of Kantorovich potentials in the transport between $\rho_t^{(1)}$ and $\rho_t^{(2)}$ for the cost $\frac{1}{p}|x - y|^p$.

Proof. We first prove the result under the assumption that $\rho^{(i)}$ is Lipschitz continuous in (t, x) and $\mathbf{v}^{(i)}$ is Lipschitz in x , uniformly in t , for $i = 1, 2$. In this case we write, for arbitrary t, t_0 , the following inequality, justified by the duality formula (Theorem 1.39):

$$\frac{1}{p} W_p^p(\rho_t^{(1)}, \rho_t^{(2)}) - \int \varphi_{t_0} \rho_t^{(1)} - \int \psi_{t_0} \rho_t^{(2)} \geq 0, \quad (5.6)$$

with equality when $t = t_0$. Choose t_0 such that $t \mapsto W_p^p(\rho_t^{(1)}, \rho_t^{(2)})$ is differentiable at $t = t_0$ and $\rho^{(i)}(t, x)$ is differentiable in t at (t_0, x) for a.e. x , and the derivative in time equals $-\nabla \cdot (\rho^{(i)}(t, x) \mathbf{v}^i(t, x))$. All these conditions are satisfied for a.e. t_0 (we use Proposition 4.3). Hence, the left-hand-side in (5.6) is differentiable at $t = t_0$ and the derivative must vanish, because of minimality at $t = t_0$. This proves

$$\frac{d}{dt} \left(\frac{1}{p} W_p^p(\rho_t^{(1)}, \rho_t^{(2)}) \right) \Big|_{t=t_0} = \int \varphi_{t_0} (\partial_t \rho^{(1)})_{t_0} + \int \psi_{t_0} (\partial_t \rho^{(2)})_{t_0},$$

and the desired formula is obtained by using $\partial_t \rho^{(i)} = -\nabla \cdot (\mathbf{v}^{(i)} \rho^{(i)})$ and integrating by parts⁶.

⁶One of the difficulties in this result is that the functions $t \mapsto \int \varphi_{t_0} \rho_t^{(1)}$ and $t \mapsto \int \psi_{t_0} \rho_t^{(2)}$ are differentiable for a.e. t , but not necessarily at $t = t_0$ (we know that for every integrand the integral is differentiable for a.e. T , and not that for a.e. t differentiability occurs for every integrand). We propose a way to overcome this difficulty via Lipschitz solutions. As an alternative approach, one can try to find the precise set of times t such that differentiability occurs for every integrand, as in [156], based on ideas from [17].

To deal with the general case, first we reduce to the case where the curves $\rho^{(i)}$ are Lipschitz in $\mathbb{W}_p(\Omega)$, by reparametrization in time. Then, we apply the regularization procedure of Section 5.3.3. In order to preserve compactness, we choose to apply the construction on the torus, and we suppose $\Omega \subset \frac{1}{2}Q$ (which is possible up to rescaling). Hence, we obtain two smoother families of curves $\rho_t^{(i,\varepsilon)}$ with their velocity fields $\mathbf{v}_t^{(i,\varepsilon)}$. For $t_0 < t_1$, the first part of the proof implies

$$\frac{1}{p} W_p^p(\rho_{t_1}^{(1,\varepsilon)}, \rho_{t_1}^{(2,\varepsilon)}) - \frac{1}{p} W_p^p(\rho_{t_0}^{(1,\varepsilon)}, \rho_{t_0}^{(2,\varepsilon)}) = \int_{t_0}^{t_1} dt \int_{\mathbb{T}^d} \nabla \varphi_t^\varepsilon \cdot E_t^{(1,\varepsilon)} + \nabla \psi_t^\varepsilon \cdot E_t^{(2,\varepsilon)},$$

where $E_t^{(i,\varepsilon)} = \rho_t^{(i,\varepsilon)} \mathbf{v}_t^{(i,\varepsilon)}$ (also set $E_t^{(i)} = \rho_t^{(i)} \mathbf{v}_t^{(i)}$) and φ_t^ε and ψ_t^ε are Kantorovich potentials in the transport from $\rho_t^{(1,\varepsilon)}$ to $\rho_t^{(2,\varepsilon)}$.

We just need to pass to the limit in the above equality. The left-hand-side trivially converges to the corresponding Wasserstein distances between the measures $\rho^{(i)}$ (note that we are on a compact space and the convolutions weakly converge to the original measure). Next, let us analyze the first part of the right-hand-side (i.e. for $i = 1$, the other part being completely analogous).

For each t we have

$$\left| \int_{\mathbb{T}^d} \nabla \varphi_t^\varepsilon \cdot E_t^{(1,\varepsilon)} \right| \leq \text{Lip}(\varphi_t^\varepsilon) \left\| \mathbf{v}_t^{(i,\varepsilon)} \right\|_{L^1(\rho_t^{(i,\varepsilon)})} \leq C \left\| \mathbf{v}_t^{(i,\varepsilon)} \right\|_{L^p(\rho_t^{(i,\varepsilon)})} \leq C \left\| \mathbf{v}_t^{(i)} \right\|_{L^p(\rho_t^{(i)})} \leq C,$$

where we used Property 5 of Proposition 5.18 and the fact that all the functions φ^ε are equi-lipschitz (see Section 1.3.2).

It is now enough to look at the limit for fixed t , since as a function of t we can apply dominated convergence. Let us set $\varphi_t^{\varepsilon\varepsilon} := \eta^\varepsilon * \varphi_t^\varepsilon$: we have

$$\int_{\mathbb{T}^d} \nabla \varphi_t^\varepsilon \cdot E_t^{(1,\varepsilon)} = \int_{\mathbb{T}^d} \nabla \varphi_t^\varepsilon \cdot (\eta^\varepsilon * E_t^{(1)}) = \int_{\mathbb{T}^d} \nabla \varphi_t^{\varepsilon\varepsilon} \cdot E_t^{(1)},$$

because of standard properties of convolutions. Thus, we only need to prove that $\nabla \varphi_t^{\varepsilon\varepsilon} \rightarrow \nabla \varphi_t$ a.e. The functions $\varphi_t^{\varepsilon\varepsilon}$ are equi-lipschitz and equi-semi-concave (see again Section 1.3.2) on a compact set and, up to translating their mean, we can apply the Ascoli Arzelà Theorem. Thus they converge up to subsequence to a function $\tilde{\varphi}$. Note that, because of semi-concavity, the convergence of the functions implies a.e. convergence of the gradients (see, for instance, Theorem 3.3.3 in [103]). We just need to prove $\nabla \tilde{\varphi} = \nabla \varphi_t$ (which would also imply that the convergence is true on the whole sequence).

Note that by Theorem 1.52 the limit of φ_t^ε must be a Kantorovich potential. Moreover, we have the general estimate

$$\begin{aligned} |(\eta^\varepsilon * f)(x) - f(x)| &\leq \int |f(x) - f(x-y)| \eta\left(\frac{y}{\varepsilon}\right) \frac{dy}{\varepsilon^d} \\ &\leq \text{Lip}(f) \int |y| \eta\left(\frac{y}{\varepsilon}\right) \frac{dy}{\varepsilon^d} = \varepsilon \text{Lip}(f) \int |z| \eta(z) dz, \end{aligned}$$

(where we used the change of variable $y = \varepsilon z$). The same estimate also works for periodic convolution, and shows $\|\varphi_t^{\varepsilon\varepsilon} - \varphi_t^\varepsilon\|_{L^\infty} \leq C \text{Lip}(\varphi^\varepsilon) \varepsilon \leq C\varepsilon \rightarrow 0$. This allows

to conclude that the limit of $\varphi_t^{\varepsilon\varepsilon}$ must be the same as that of φ_t^ε , i.e. a Kantorovich potential. Even if the Kantorovich potential is not necessarily unique, its gradient is unique, and we deduce $\nabla \varphi_t^{\varepsilon\varepsilon} \rightarrow \nabla \varphi_t$. $\square$

Note that in the above proof we intensively used the compactness of the domain Ω (and of the torus: the same proof performed with regularization via measures on $\mathbb{R}^d$ seems delicate because of the lack of bounds on $\nabla \varphi_t^{\varepsilon\varepsilon}$). The assumption that the measures are absolutely continuous is used to take advantage of the a.e. convergence of the gradients to a unique limit (the gradient of the limit potential). Without this, one could have problems in recovering full convergence of the sequence.

As a corollary of the previous result we obtain the following useful formula.

Corollary 5.25. *Under the same assumptions of Theorem 5.24, we have,*

$$\frac{d}{dt} \left(\frac{1}{p} W_p^p(\rho_t^{(1)}, \rho_t^{(2)}) \right) = \int_{\Omega} (x - T_t(x)) \cdot \left(\mathbf{v}_t^{(1)}(x) - \mathbf{v}_t^{(2)}(T_t(x)) \right) \rho_t^{(1)}(x) dx,$$

where T_t is the optimal transport map from $\rho_t^{(1)}$ to $\rho_t^{(2)}$ for the cost $\frac{1}{p}|x-y|^p$.

Proof. We just need to re-write the conclusion of Theorem 5.24, using $\nabla \varphi_t(x) = x - T_t(x)$ and $\nabla \psi_t(y) = y - S_t(y)$, where S_t is the optimal transport from $\rho_t^{(2)}$ to $\rho_t^{(1)}$. We use

$$\int (y - S_t(y)) \cdot \mathbf{v}_t^{(2)}(y) d\rho_t^{(2)}(y) = \int (T_t(x) - x) \cdot \mathbf{v}_t^{(2)}(T_t(x)) d\rho_t^{(1)}(x),$$

which is a consequence of $(T_t)_\# \rho_t^{(1)} = \rho_t^{(2)}$ and $S_t = T_t^{-1}$. Hence, the formula given by Theorem 5.24 provides

$$\begin{aligned} \frac{d}{dt} \left(\frac{1}{p} W_p^p(\rho_t^{(1)}, \rho_t^{(2)}) \right) \\ = \int_{\Omega} (x - T_t(x)) \cdot \mathbf{v}_t^{(1)}(x) d\rho_t^{(1)}(x) + \int_{\Omega} (T_t(x) - x) \cdot \mathbf{v}_t^{(2)}(T_t(x)) d\rho_t^{(1)}(x), \end{aligned}$$

which gives the claim. $\square$

Some examples of application of the above computation to the uniqueness and the properties of the solution to some evolution PDEs are presented in the Exercise section (**Ex(34)**, **Ex(66)**, **Ex(67)**, **Ex(68)**, **Ex(69)**).

Here, we will only give an example of application to the uniqueness for the continuity equation with given vector field.

Proposition 5.26. *Let $\mathbf{v}_t : \Omega \rightarrow \mathbb{R}^d$ be a family of vector fields on a compact smooth domain Ω , parametrized in time, satisfying a one-sided Lipschitz condition $(\mathbf{v}_t(x) - \mathbf{v}_t(y)) \cdot (x - y) \leq C|x - y|^2$ for a fixed constant C , not depending on x, y nor t . Suppose also that $\mathbf{v}_t \cdot \mathbf{n} = 0$ on $\partial\Omega$ and consider the equation $\partial_t \rho_t + \nabla \cdot (\rho_t \mathbf{v}_t) = 0$ with given initial datum ρ_0 . Then there is at most one solution ρ_t among AC curves in $\mathbb{W}_2(\Omega)$ which are absolutely continuous for a.e. t .*

Proof. Consider two solutions $\rho_t^{(1)}, \rho_t^{(2)}$ and compute the derivative in time of $W_2^2(\rho_t^{(1)}, \rho_t^{(2)})$. Using Corollary 5.25, we have

$$\begin{aligned} \frac{d}{dt} \left(\frac{1}{2} W_2^2(\rho_t^{(1)}, \rho_t^{(2)}) \right) &= \int_{\Omega} (x - T(x)) \cdot (\mathbf{v}_t(x) - \mathbf{v}_t(T(x))) \rho_t^{(1)} dx \\ &\leq C \int_{\Omega} |x - T(x)|^2 \rho_t^{(1)} dx = C W_2^2(\rho_t^{(1)}, \rho_t^{(2)}). \end{aligned}$$

This allows to apply Gronwall's lemma and deduce $W_2^2(\rho_t^{(1)}, \rho_t^{(2)}) = 0$ from the fact that the same distance vanishes at $t = 0$. $\square$ $\square$

Note that the above proof could also be used to prove uniqueness in the continuity equation, and applied to Theorem 4.4.

5.4 Constant speed geodesics in $\mathbb{W}_p$

We will see in this section how constant speed geodesics in W_p are related to optimal transport maps. Before doing so, we recall the main facts about geodesics in metric spaces.

Box 5.2. – Good to know! – Constant speed geodesics in general metric spaces

First of all, let us define the length of a curve ω in a general metric space (X, d) .

Definition - For a curve $\omega : [0, 1] \rightarrow X$, let us define

$$\text{Length}(\omega) := \sup \left\{ \sum_{k=0}^{n-1} d(\omega(t_k), \omega(t_{k+1})) : n \geq 1, 0 = t_0 < t_1 < \dots < t_n = 1 \right\}.$$

Note that the same definition could be given for functions defined on $[0, 1]$ and valued in X , not necessarily continuous. The functions ω with $\text{Length}(\omega) < +\infty$ are exactly those which have bounded variation in the sense of the BV functions 1D (see Box 4.4 in Section 4.2.4). It is easy to see that all curves $\omega \in \text{AC}(X)$ satisfy $\text{Length}(\omega) \leq \int_0^1 g(t) dt < +\infty$.

Proposition - For any curve $\omega \in \text{AC}(X)$ we have

$$\text{Length}(\omega) = \int_0^1 |\omega'(t)| dt.$$

We collect now some more definitions.

Definition - A curve $\omega : [0, 1] \rightarrow X$ is said to be a geodesic between x_0 and $x_1 \in X$ if it minimizes the length among all curves such that $\omega(0) = x_0$ and $\omega(1) = x_1$.

A space (X, d) is said to be a *length space* if it holds

$$d(x, y) = \inf \{ \text{Length}(\omega) : \omega \in \text{AC}(X), \omega(0) = x, \omega(1) = y \}.$$

A space (X, d) is said to be a *geodesic space* if it holds

$$d(x, y) = \min \{ \text{Length}(\omega) : \omega \in \text{AC}(X), \omega(0) = x, \omega(1) = y \},$$